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6

Debug and Fix DFT DRC Violations

Learning Objectives

After completing this lab, you should be able to:

- Debug dft_drc problems with clocks and resets
- Propose a manual solution that could be implemented by you or by another designer
- Use AutoFix to implement a fix



Lab Duration:
45 minutes

Lab 6

Overview

All files for this lab except for designs and libraries are located in the directory **lab6_fixing**.

Directory Structure

lab6_fixing	Current working directory
analyzed	Intermediate acs_read_hdl files
logs	Session log files
unmapped	Unmapped protocol
mapped	Gate-level netlist
mapped_scan	Gate-level post scan netlist
reports	DFTC reports
tmax	Files for downstream tools
scripts	Constraint and run scripts
../ref/rtl/RISC_CORE_nodft/vhdl	Design files
../ref/db	Library files
../.solutions	Solution files

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, consult the back portion of this lab for help.

Instructions

During this lab you will debug Uncontrollable Clock and Reset DFT problems using the Design Vision GUI, propose a fix and implement a fix using AutoFix.

Task 1. Read in Mapped Design

1. Change directories to the project directory **lab6_fixing**.
2. Invoke DC in the GUI mode.

```
UNIX% dc_shell -gui
```

3. Read in the mapped design. Remember file name completion!

```
source scripts/4read_gate_and_protocol.tcl
```

清除design.
读入dc.link ②
设 set-dft-signal
create-test-protocol
dft-div 检查

Task 2. Debug Uncontrollable Clocks and Resets

1. From information in the log window, answer the following:

Question 1. How many violations are there?
What type of violations are they?

print var 可显示 dcd
环境变量。

```
5 TOPOLOGY VIOLATIONS
  5 Unconnected input pin violations (TEST-332)

346 PRE-DFT VIOLATIONS
  258 Uncontrollable clock input of flip-flop violations (D1)
   5 DFF set/reset line not controlled violations (D2)
  83 DFF set/reset line not controlled violations (D3)
```

2. In the Violation Browser, analyze the first D1 violation and you should observe a schematic similar to the one shown below.

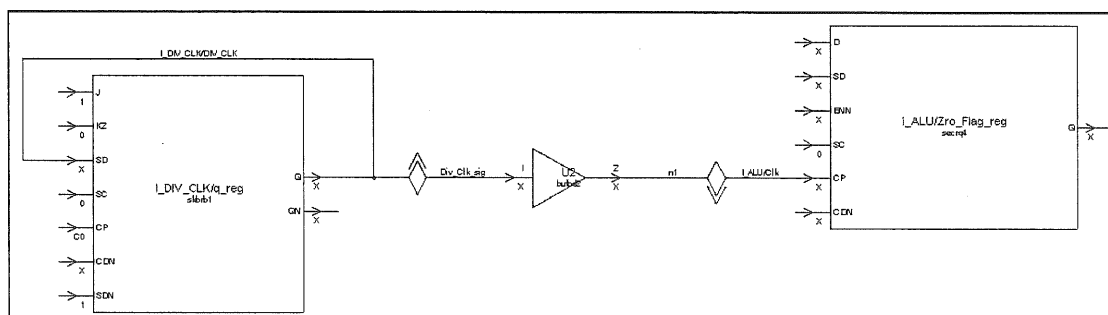


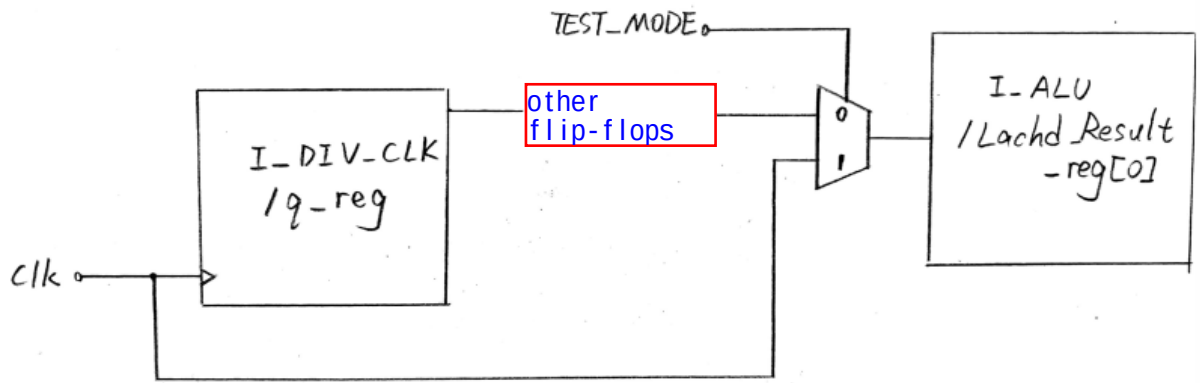
Figure 1.

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Question 2. What is the netlist name for the flip-flop on the left, the cause for the D1 violation?

I_DIV_CLK/q_reg

Question 3. Draw a schematic “fix” for this uncontrollable clock problem.



3. Analyze some of the other D1 violations. Select the Violation Browser tab, then select some of the other D1 violations and examine the schematic.

Question 4. What can you conclude about the D1 violations in the RISC_CORE design?

All comes from the same problems: Clk should be set_dft_signal ...

Analyze one of the D3 violations in the same manner as you did the D1 violations. To expand the second X input to I_RST/q2_reg, double click on the D input.

Question 5. Based on your analysis draw a schematic “fix” for the uncontrollable reset problem.

与上图解决方案相似，将I_DIV-CLK/q_reg替换为I_RST/q2_reg，将Clk信号替换为Reset信号并连接到reset端即可。

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4. Select some of the other D3 violations and analyze them.

Question 6. What can you conclude about the D3 violations in the RISC_CORE design?

All comes from the same problems: Reset should be set_dft_signal ...

Task 3. Autofix The D1, D2, and D3 violations

You have now analyzed both the uncontrollable clock and uncontrollable reset DFT violations in the RISC_CORE design. Your next task is to fix the violations. As you learned in the lecture there are three choices.

- 1) You could re-code the RTL design and check the design once again using the unmapped DFTflow. When you have no more violations you can continue on to the mapped flow.
- 2) You could draw the solution and have someone else re-code the design for you.
- 3) You can use the AutoFix capability of DFTC.



Question 7. What kind of DFT problems can Autofix solve?

pg-1 3分

1.uncontrollable clocks/resets/sets signals
2.internal tristates signals
3.bidirectional signals

1. The 5preview_dft.tcl file is setup to **source** the autofix.tcl script **after** sourcing the **settings_insert_dft.tcl** file.

Set up to run AutoFix by modifying a script called **scripts/autofix.tcl**.

Utilize the lecture material and the man pages to modify the script called **autofix.tcl** in the **scripts/** directory. Enable AutoFix to fix uncontrolled clocks, uncontrolled resets, and uncontrolled sets.

Here is a list of commands related to AutoFix to get you started:

- **set_dft_configuration**
- **set_autofix_configuration**
- **set_dft_signal**

2. Preview what AutoFix will add by running step 5 of the Mapped Flow.

```
source -e -v scripts/5preview_dft.tcl
```

报error 缺少type-
退出重新进可以了

3. Evaluate the previewed AutoFix results.

TEST MODE: no hookup pin 什么意思

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```
***** Test Point Plan Report *****
Total number of test points : 2
Number of Autofix test points: 2
Number of Wrapper test points: 0
Number of test modes : 1
Number of test point enables : 0
Number of data sources : 2
Number of data sinks : 0
*****
```

```
Dft signals:
TestMode: TEST_MODE (no hookup pin)
TestData: Clk (no hookup pin)
TestData: Reset (no hookup pin)
```

Take a careful look at the results of running the `preview_dft` command. Notice the sections of the report titled **Test Point Plan Report**, and **DFT signals**. These sections relate directly to the AutoFix portion of the flow.

Compare your results carefully with those in the Answers / Solutions section—it is easy to make a mistake and not realize it.

Question 8. How many AutoFix test points are there? Why?

2.
the one comes from uncontrollable clk, the other one comes from uncontrollable reset (Q3)

Question 9. What do you think would happen if a register had an uncontrolled reset **and** uncontrolled set and the same top-level signal was specified as the TestData?

can not shift value correctly

4. Perform both AutoFix and scan insertion by running step 6 of the DFT flow.

```
source -e -v scripts/6insert_dft.tcl
```

insert_dft.
15 ft coverage.

5. Look at the results of running `dft_drc -coverage_estimate`. *会报 violation*

Question 10. Were there any violations reported? What was the fault coverage?

```
DRC Report
Total violations: 0
```

```
Uncollapsed Stuck Fault Summary Report
-----
fault class      code  #faults
-----
Detected         DT    10408
Possibly detected PT     0
Undetectable     UD     37
ATPG untestable  AU     7
Not detected     ND     4
-----
total faults          10456
test coverage         99.89%
```

Question 11. What kind of fault coverage would you expect to get, if you had not AutoFixed the design?

extreme low...

```
Uncollapsed Stuck Fault Summary Report
-----
fault class      code  #faults
-----
Detected         DT     39
Possibly detected PT     0
Undetectable     UD    797
ATPG untestable  AU   9636
Not detected     ND     0
-----
total faults          10472
test coverage         0.40%
```

6. Try it and compare coverages.

7. Exit DFTC.

Congratulations! You have completed this lab.